

NegletFix: measured VR rehabilitation for chronic left homonymous hemianopia

Prepared for Dr. Thierry Jacquemin. The goal is to explain what Eric is building, what evidence it is based on, how progress will be measured, and what medical feedback would be useful before starting a structured training block.

Prepared

May 26, 2026
Patient: Eric Lespagnon
Project: NegletFix

Purpose of the appointment discussion

This is not a request for a prescription or a claim that the system already works. It is a measured rehabilitation project for a chronic visual-field deficit after stroke. I would like the doctor's view on medical safety, useful clinical measurements, and whether the assumptions make sense from a physician's point of view.

1. One-Minute Summary

Condition

Left homonymous hemianopia after a right PCA stroke in July 2021, with persistent left visual-field loss and severe left contrast deficit.

Project

NegletFix uses VR to deliver repeated audiovisual training in the affected left field, while measuring contrast sensitivity and functional change over time.

Doctor Input Needed

Confirm safety, suggest clinical outcome measures, review fatigue/medication/lab issues, and flag any contraindication before disciplined training starts.

The clean medical framing

NegletFix is best understood as an **N-of-1, measurement-first rehabilitation protocol**. The first goal is not to prove a cure. The first goal is to run a repeatable, low-risk training block and objectively determine whether Eric's left hemifield detection, contrast sensitivity, reaction time, or daily function changes beyond measurement noise.

2. Eric's Medical Context

Item	Current understanding
Primary diagnosis	Left homonymous hemianopia after right posterior cerebral artery stroke, July 2021.
Imaging history	Large area of encephalomalacia in the right occipital lobe.
Main daily symptoms	Complete left visual-field blindness, severe left-side contrast deficit, persistent global dimness / "gray overlay", screen fatigue, and functional difficulty with left-side awareness.
What this is not	It is not only comfort, brightness preference, or a UI problem. The left-field deficit is objective and matches the stroke location.
Rehabilitation target	Improve measurable visual function or daily compensation enough to matter. Full visual-field restoration is not assumed.

3. Baseline Measurement Already Obtained

NegletFix includes a modified Pelli-Robson-style contrast sensitivity test. A critical positioning bug was fixed before the valid baseline below: letters now appear in the intended hemifield while Eric fixates a central cross.

Hemifield	LogCS result	Approx. threshold	Interpretation
Central	~1.05+	~9 percent contrast	Usable central baseline; useful as a display/fatigue calibration check.
Right / intact field	2.25	~0.6 percent contrast	Excellent; maxed out the current test.
Left / affected field	0.00	100 percent contrast not detected	Severe deficit; pure black-on-gray letters were not identified in the affected hemifield.

Asymmetry: 2.25 LogCS, far above the usual clinically meaningful change threshold of about 0.30 LogCS. The plan is to compare future results against this baseline, not against population norms from a different instrument.

Measurement caveat

The NegletFix contrast test is currently best used for **within-device longitudinal tracking**. LCD/OLED contrast measurements can differ from calibrated printed charts. That is acceptable for this project if the same device, brightness, distance, and fixation rules are kept stable.

4. Scientific Rationale

The working hypothesis is that even when the primary visual cortex pathway is damaged, some residual visual processing can be trained through alternate pathways and repeated task practice.

Concept	Why it matters for NegletFix	Confidence
Two visual pathways	The V1/geniculostriate pathway is damaged for the left field, but retinotectal and extrastriate pathways may still support orienting, motion, and partial detection.	high anatomy
Audiovisual pairing	Spatially and temporally congruent sound plus visual cues can strengthen detection in weak visual fields; this is the core of the training.	moderate clinical
Chronic stroke reality	Eric is almost 5 years post-stroke. Recovery is still possible in some chronic patients, but effect sizes are expected to be modest and variable.	mixed
Daily-function transfer	The most important question is whether lab gains transfer to reading, walking, screens, cooking, navigation, and fatigue. This must be tracked directly.	unknown

5. Research Findings We Are Using

Evidence branch	What it suggests	How NegletFix uses it
Daibert-Nido et al. 2021	A small pediatric hemianopia pilot reported visual-field and contrast-sensitivity gains after home immersive audiovisual training.	Useful proof-of-concept, but not enough by itself because it was not a powered adult-stroke trial.
Misawa / Daibert-Nido 2024	A home telerehabilitation version using Meta Quest 2/Pro was feasible in pediatric hemianopia.	Supports the practical idea that a Quest-based home protocol can be usable, monitored, and repeated.
Alharshan / Alwashmi 2026	Fifteen subacute/chronic stroke patients with HH completed 30 min/day, 5 days/week, for 6 weeks, with behavioral and imaging changes reported.	This is the closest direct dose model for Eric: 30 sessions over 6 weeks, about 15 total hours.
Namgung et al. 2025	A randomized clinical trial of digital visual perceptual learning in post-stroke visual field defects found measurable visual-field improvements.	Supports the broader premise that structured digital training can produce measurable chronic post-stroke visual gains in some patients.

Evidence branch	What it suggests	How NegletFix uses it
Yang / Saionz / Huxlin line	Chronic V1-damage patients can improve, but contrast sensitivity remains fundamentally limited and not everyone responds.	Sets realistic expectations: a clinically meaningful gain would be valuable, but full normalization is not the target.
ESO 2025 visual impairment guideline	Standardized screening and access to vision rehabilitation after stroke are important; the evidence base is still uneven.	Supports asking for formal measurement/referral rather than relying only on self-report.

6. Planned NegletFix Protocol

Proposed first real training block
30 minutes per day, 5 days per week, for 6 weeks if tolerated. This mirrors the closest adult-stroke audiovisual training dose found in the recent literature. Start conservatively if VR fatigue or nausea appears.

Protocol element	Implementation	Reason
Training stimulus	Co-localized visual target plus auditory cue in VR, primarily presented toward the affected left field.	Uses multisensory pairing to help weak visual signals become behaviorally useful.
Difficulty progression	Starts near the scotoma border and adjusts contrast/eccentricity based on performance.	Severe left baseline means starting too far into the blind field would produce only failure and fatigue.
Feedback	Open-loop task feedback first. EEG is not required for the first protocol.	Keeps the first training block simpler and testable.
EEG / Muse layer	Exploratory only. Validate signal quality before using it as a training gate.	Consumer EEG evidence is mixed; it should not be the foundation of the first medical discussion.
Data logging	Each trial logs eccentricity, hemifield, contrast, reaction time, hit/miss, and audiovisual timing.	Allows the project to learn from actual data rather than memory or impression.

7. What Is Already Built

Component	Status	Doctor relevance
Contrast sensitivity test	Operational; valid baseline obtained after bug fix.	Gives objective pre/post tracking.
Audiovisual training module	Scaffolded in Unity; Editor smoke test pending.	Main rehabilitation method to discuss.
Quest VR setup	Quest 2 developer mode working; OpenXR configured.	Home delivery is technically feasible.
Data logger	CSV trial logging wired.	Supports objective review and later medical summary.
Muse / EEG	Code exists, real-world signal validation not done.	Should remain a future layer, not the clinical core today.

8. Outcome Measures To Discuss

Measure	How to use it	Why it matters
NegletFix contrast sensitivity	Central + left quick-check every 5 sessions; full central/right/left reassessment at baseline and post-program.	Primary project measurement.
Formal visual field test	Humphrey, Goldmann, or local equivalent if available.	Clinically recognizable external anchor for the doctor file.

Measure	How to use it	Why it matters
Reaction time and hit rate	From the VR task logs at specific eccentricities.	May improve before conscious contrast detection improves.
Daily-function tracker	Short weekly ratings: screen fatigue, cooking, navigation, left-side collisions/surprises, reading, evening exhaustion.	Prevents the project from chasing a lab number that does not help daily life.
Fatigue / sleep / mood / medication review	Review during routine care and prescription renewal.	Training requires vigilance; sedation, poor sleep, depression, or uncontrolled vascular risk can bury any signal.

9. Questions For Dr. Jacquemin

- 1. Does the medical framing make sense?** Chronic left homonymous hemianopia after right PCA stroke, with a measured contrast-sensitivity asymmetry.
- 2. Is there any reason VR audiovisual training would be unsafe for me?** Especially seizure history, migraines, vestibular symptoms, severe fatigue, blood pressure, or photosensitivity.
- 3. Should I get formal visual-field testing before the training block?** If yes, what test and through whom: ophthalmology, neuro-ophthalmology, orthoptist, or rehabilitation clinic?
- 4. Should we distinguish hemianopia from neglect more formally?** NegletFix started from "neglect" language, but my documented problem is homonymous hemianopia. If attention/neglect is also present, that changes the interpretation.
- 5. What blood work or prescription review matters for training?** Lipids, HbA1c/glucose, CBC, CMP/electrolytes, B12, vitamin D, TSH, blood pressure, sleep, fatigue, and medication side effects may affect rehabilitation capacity.
- 6. Would you be willing to review pre/post results later?** The useful doctor role may be to sanity-check whether the data look meaningful or only noise.
- 7. Do you see any specialist referral that would make this safer or more rigorous?** Neuro-ophthalmology, OT/orthoptics, stroke rehab, or visual rehabilitation specialist.

10. Safety Boundaries

Not a substitute

This does not replace medical care, vascular prevention, eye exams, formal rehab, or physician-directed treatment.

Not driving clearance

Any functional improvement in VR is not permission to drive. Driving decisions require formal legal and medical assessment.

Stop rules

Pause training if it provokes major headache, vertigo, nausea, unusual neurological symptoms, visual distress, or marked fatigue.

11. Side Question: Hyperbaric Oxygen

Why this is only a side note

Eric remembered hyperbaric oxygen therapy from a recovery podcast and noticed that the practice may have a hyperbaric module. After a quick evidence check, this does **not** look like the main NegletFix topic.

HBOT is real medicine, but stroke recovery is not a standard accepted indication. Acute ischemic stroke guidelines do not recommend HBOT except for stroke caused by air embolization. Chronic-stroke studies exist, but I did not find strong direct evidence for chronic post-PCA homonymous hemianopia or left-field contrast restoration.

If Dr. Jacquemin thinks HBOT is relevant anyway, the safe framing would be: off-label adjunct, clear contraindication screening, and pre/post measurement. It should not replace the core NegletFix protocol.

HBOT discussion point	Question for doctor
Relevance	Do you see any plausible role in my chronic visual-field condition, or would you consider it unrelated/unproven?
Safety	Any contraindications in my history: ears/sinuses, lungs, seizure risk, eye history, glucose, medications, blood pressure?
Measurement	If ever considered, should it wait until after a clean baseline NegletFix block so we do not confuse what caused any change?

12. Suggested Way To Explain It Out Loud

"I built this project because my left visual field is still functionally blind after the right PCA stroke. We now have a measured baseline: right field is excellent, left field is at the floor. The idea is to run a controlled VR audiovisual training block, not as a miracle cure, but as a measurable rehab experiment. I want your opinion on whether it is safe, what clinical measurements would make it more serious, and whether anything in my prescriptions, fatigue, blood work, or stroke prevention plan could affect the training."

13. Sources Checked

- Daibert-Nido et al. 2021, audiovisual 3D immersive stimulation in hemianopia, PubMed: <https://pubmed.ncbi.nlm.nih.gov/34106907/>
- Daibert-Nido et al. 2021, full Frontiers article: <https://www.frontiersin.org/articles/10.3389/fneur.2021.680211/full>
- Misawa et al. 2024, EClinicalMedicine telerehabilitation feasibility trial, PubMed: <https://pubmed.ncbi.nlm.nih.gov/39687429/>
- Alharshan / Alwashmi et al. 2026, immersive audiovisual training in post-stroke hemianopia, PubMed: <https://pubmed.ncbi.nlm.nih.gov/41421499/>
- Alharshan / Alwashmi et al. 2026, NeuroImage article: <https://www.sciencedirect.com/science/article/pii/S105381192500672X>
- Namgung et al. 2025, JAMA Network Open randomized clinical trial of digital visual perceptual learning: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12090032/>
- European Stroke Organisation 2025 guideline on visual impairment in stroke: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12098360/>
- Yang et al. 2023, contrast sensitivity after V1 damage: <https://www.medrxiv.org/content/10.1101/2023.08.31.23294827v1>
- Treves et al. 2025, consumer-grade neurofeedback meta-analysis: <https://pubmed.ncbi.nlm.nih.gov/40246295/>
- AHA/ASA acute ischemic stroke guideline slide set, HBOT recommendation context: <https://professional.heart.org/>
- Cochrane evidence summary on HBOT for acute ischemic stroke: https://www.cochrane.org/evidence/CD004954_hyperbaric-oxygen-therapy-treating-people-acute-ischaemic-stroke
- UHMS accepted hyperbaric oxygen indications: <https://www.uhms.org/resources/featured-resources/hbo-indications.html>

This document is a physician discussion aid. It does not recommend starting, stopping, or changing medication; it does not recommend HBOT; and it does not claim NegletFix is clinically proven for Eric. The intended next step is medical review plus a cautious, measured training block.